

PPO with Human Style Coaching Rewards on LunarLanderContinuous-v3



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Project Motivation

- Standard PPO learns only from environment rewards
- Human feedback is common in real-world learning
- Hypothesis: Human style coaching rewards leads to faster/more stable learning

Tools and Frameworks Used

- **Gymnasium** : LunarLanderContinuous-v3 environment
- **PyTorch** : PPO neural networks
- **Stable-Baselines3** structure
- **Matplotlib / Numpy** : learning curve analysis
- **Custom wrapper** for coaching rewards

Baseline PPO Model Details

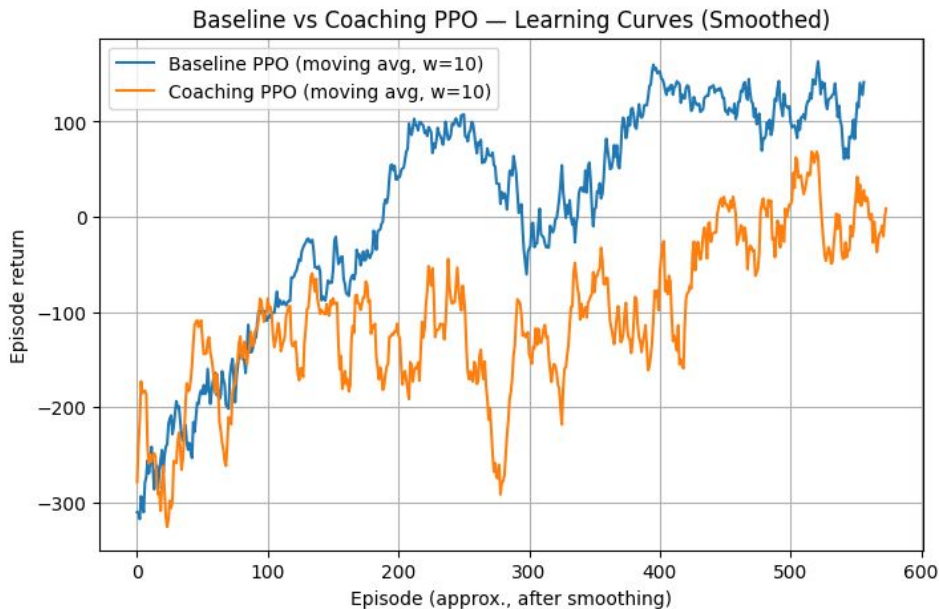
- PPO architecture
 - Actor + Critic (MLP)
 - Two Hidden Layers (e.g. 64-64)
- Trained for 300,000 timesteps
- Uses only environment reward
- Standard PPO features: clipping GAE, entropy bonus

Human Style Coaching Model: What Changed

- Added coaching reward components:
 - Bonus for staying near center
 - Bonus for staying upright
 - Penalty for excessive thruster use
- Coaching reward added in addition to original reward
- Same PPO architecture & hyperparameters to ensure fairness

Learning Curve Comparison

- Coaching PPO Learns lower.
- More variance and Instability
- Baseline PPO consistently achieves returns during training.



Policy Evaluation Results

Baseline PPO:

- Mean return = 192.81
- Std = 90.52

Coaching PPO:

- Mean return = 225.75
- Std = 39.37

- Final policy evaluation over 10 episodes

Limitations & Next Steps

Limitations:

- Training instability from coaching rewards
- Coaching signals may conflict with environment rewards
- Small dataset of 10 evaluation episodes
- Only tested on one environment

Next Steps:

- Tune or scale coaching rewards
- Use adaptive or decaying shaping
- Expand to more complex environments

Conclusion

- Coaching made training noisier and less stable
- But produced a better and more consistent final policy
- Highlights the importance of reward design
- Coaching rewards influence exploration, not just performance,
- Human guidance can help, but needs to be applied carefully

References

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